

LinkedIn: <https://linkedin.com/in/kojikanao>GitHub: <https://github.com/koji>Portfolio: <https://koji.thedev.id/>

TECHNICAL SKILLS

- Proficient:** JavaScript, React.js, Redux, TypeScript, Node.js, Webpack, Python, openFrameworks, Processing, PostgreSQL, Git, Electron, Jest, testing library/react, Storybook, Vite, yarn, Vitest, Jotai, GitHub Actions, Cypress, pnpm, bun, CSS Module, Chakra UI, Cursor, Claude Code
- Exposure:** Deno, Gatsbyjs, Nextjs, GLSL, TouchDesigner, OpenCV, Docker, CircleCI, Sketch, rollup.js, three.js, bash, Tailwind CSS

PROFESSIONAL EXPERIENCE

Opentrons | Software Engineer | New York

March 2022 - Present

- Led UI modernization efforts through a migration to CSS Modules, accessibility improvements, and UI component refactoring.
- Leading AI enablement initiatives by building and maintaining shared harnesses and subagents, accelerating AI adoption and engineering productivity across teams.
- Collaborated with designers to build and launch a UI component library using React and Storybook, which improved component reusability, streamlined code reviews, and increased development speed by 50%
- Led development of an Electron and React-based on-device display application, driving key UX features and enabling cross-team development through a touchscreen-compatible platform.
- Developed and deployed an internal automation tool to automatically post robot usage statistics to Slack, removing repetitive manual tasks and saving roughly 30 minutes per engineer per month
- Leading the frontend architecture and development of an AI-powered robotics application, driving user experience and technical implementation for a solution showcased at SLAS and widely recognized by attendees.
- Spearheaded the adoption of Sentry, fixed critical production issues, and improved the application's error-free session rate to 99.5%
- Reduced CI wait times by 9 minutes from 15 minutes and cut local environment setup time by 80% through migrating from yarn to pnpm

Pivotal Advisors | Director of Technology | New York

November 2020 - December 2021

- Analyzed client data and rebuilt the website. Also designed and developed a system to automate company operations, resulting in a 15% increase in inquiries from potential customers.

Ubiquiti Networks, Inc | Software Engineer(R&D) | New York

July 2018 - July 2020

- Using React.js, TypeScript, three.js, Redux, Yarn, styled-components, react-draggable, and PostgreSQL, built an IoT hub system that displays 3D models in the web browser, allows placement of IoT devices within the 3D model, and enables users to visually control devices, such as checking security camera angles and visualizing router signals. Subsequently, converted the frontend application into an npm package, enabling overseas development teams to easily integrate the 3D application into their apps, and led the project as the lead engineer
- Optimized the npm package to shorten build time with Webpack, and I could reduce the duration by over 90%
- Built and launched UniFi-Network-AR to simplify IT maintenance tasks using C#, Unity, JavaScript, and Node.js for beta testers
- Developed a machine learning application to detect rooms from an image using Python, Keras, and Beautiful Soup, Selenium, correctly 150-floor plans from Matterport Gallery, processing and predicting 5-floor plans from 3D models

New York University | Researcher | New York

August 2018 - July 2019

- Provided comprehensive mentorship to students, offering technical guidance on projects and strategic support for their career development and job search
- Developed a low-cost WiFi/Bluetooth signal-sniffing device with Raspberry Pi, Aircrack-ng, Bluez, and Python, contributing to a proof of concept that became a finalist

N-GEMS, Inc | Co-Founder/Lead Engineer | New York

February 2017 - January 2018

- Built an N-GEMS application using Node.js and Raspberry Pi to capture and visualize signals, and presented a demo-ready prototype, resulting in selection for the NYC Media Lab Combine cohort
- Conducted research on low-cost WiFi/Bluetooth signal sniffing devices using Raspberry Pi, Aircrack-ng, Bluez, and Python, identified the key technologies needed for the N-Gems service, and led the proof of concept. As a result, the team became a finalist in a startup competition and successfully secured small-scale support from investors

SCSK, Inc | System Engineer | Japan

April 2010 - October 2013

- Developed and maintained power plants' control systems with C, C++, Java, and Unix OS, helping maintain 99% uptime
- Built tools for the project team and manager using VBA, reducing operation time from 5 minutes to 1 minute

tayorie | founding Software Engineer (contract) | Remote

August 2025 - December 2025

- Help the team to build the brand new product of LINE application which is written Typescript, [Next.js](#), Chakra UI and Firebase

EDUCATION

M.A., New York University - Interactive Telecommunication Program

May 2018

M.A., Kobe University - Econometrics

March 2010

B.S., Tokyo University of Science - Operations Research

March 2008

PROJECT WORK

Hyouji(表示) | Developer | [npm URL](#) | [GitHub](#)

2025 - present

Hyouji is a simple CLI tool to create/delete labels with GitHub Labels API. Now available as a global npm package with persistent configuration storage

Opentrons Document MCP Server | Developer | [npm URL](#) | [GitHub](#)

2025 - present

A MCP server for Opentrons product such as OT-2, Opentrons Flex, Protocol Designer and Python API

react-nes-components | Developer | [Deploy URL](#) | [GitHub](#)

2019 - present

react-nes-components is a React UI component that allows developers to use NES.css with React.js